

UNIT 2

TABLETS



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Introduction

- **Tablets** are solid DFs prepared by **compression** of medicaments with a variety of excipients.
- Most of the time, it is intended for oral use and usually swallowed with water.
- Oral tablets may be swallowed *whole*, after being *chewed*, or after being *dissolved* in water.
- Oral tablets are designed to release the medicament with in the GIT for absorption into the circulation, or more rarely for a local effect.
- It varies in shape, size, color, hardness, thickness, weight, disintegration, and dissolution characteristics

Introduction...

Advantages of tablets

- ✓ *Compact* dosage forms
- ✓ *Convenient* and safe way of drug administration
- ✓ Preparation procedure enables *accurate dosing* of the drug
- ✓ Easier to **package/store** and transport
- ✓ Easier to **mask** the taste of bitter drugs
- ✓ Generally **inexpensive** dosage form

Introduction...

Disadvantages of tablets

- **Multistep** production
- Not suitable for drugs with poor *solubility* and *absorption*
- Some drugs may cause *local irritant* effects
- Not suitable for drugs with *poor compressibility*
- Administration of tablets may be problematic in **children** and **elderly** patients

Introduction...

DESIRED CHARACTERISTICS OF TABLETS

- Should be elegant product having its own identity, free from defects as chips, cracks, discoloration.
- Should have strength to withstand mechanical shock during production, packaging or shipping.
- Must be able to release the medicament in a predictable and reproducible manner.
- Should be chemically, physically and microbiologically stable.

Tablet Excipients

- ❑ Conventional oral tablets for ingestion usually contain the same classes of components in addition to the active ingredients, which are one or more agents functioning as:-
 - **Diluents**
 - **Binders**
 - **Disintegrants**
 - **Lubricants**
 - **Colorants, flavourants and sweeteners**

Tablet Excipients...

CRITERIA FOR USE AS EXCIPIENTS

- Tablet excipients must meet certain criteria in the formulation. These include the following:-
 - It must be *non-toxic* and *acceptable* to the regulatory agencies in all countries where the product is to be marketed.
 - **There cost must be acceptably low.**

Tablet Excipients...

- Must not be contraindicated by themselves
- Must be physiologically inert
- Must be *physically* and *chemically stable* by themselves and in combination with the drug and other tablet components
- Must be free of any **unacceptable microbiologic load**
- Must have **no deleterious effect on the BA** of the drug in the product

Diluents

- They are fillers designed to make up the **required bulk** of the tablet when the drug dosage itself is inadequate to do so.
- Secondary reasons in using diluents in the tablet formulation include providing better tablet properties such as improved **cohesion** or **promote flow**.
- In potent drugs diluents increase the bulk → *weighed* easily and *divided* conveniently
- Different substances, mainly *carbohydrates* and *inorganic salts* are used as fillers
- **Examples:-** lactose, mannitol, sorbitol, sucrose, starch, cellulose derivatives (e.g., MCC (Avicel[®])), Dicalcium phosphate dihydrate,

Binders

- These materials are added either **dry** or **in liquid form** during wet granulation to form ***granules*** or promote ***cohesive compacts*** for directly compressed tablets.
- Promote adhesion of particles to enable a granulation to form and to maintain mechanical ***strength*** of tablet
- Best (effective) cohesion achieved with **solution binders** and is most common way of incorporation
- Both types included in formulation at relatively low concentration (2-10 %w/w)
- **Examples:-** acacia, tragacanth, gelatin, glucose, starch, PVP, sodium alginate, cellulose derivatives.....

Disintegrants

- Compact tablet obtained by compression may require hours to disintegrate and dissolve.
- To ensure its *break up* in GIT, to produce complete *dissolution* and bioavailability (BA), disintegrants are included
- **Disintegrant** is added to most tablet formulations to facilitate a *break up* or *disintegration* of the tablet when it contacts with water in the GIT.
- It may function by drawing water into the tablet, swelling, and causing the tablet to burst apart.
- **Examples:-** starch, cellulose, alginate, PVP.....

Disintegrants...

- The best and fastest acting disintegrants **hydrate** and **swell** in the acid environment of stomach (pH 1 to 3)
- Mechanism is the reverse of binders → increase the *surface area* so that the dissolution rate will increase
- Starch (traditional disintegrant -up to 10%)
- Most **effective** disintegrants act by swelling (swell dramatically during water uptake) → modified starches & cellulose → 1-5% w/w.
- Disintegrants may be used:
 - *Intragranularly*
 - *Extragranularly*
 - *In both forms*

Lubricants, Antiadherants and Glidants

- These three classes of materials are typically described together because they have overlapping functions.
- A material that is primarily described as an antiadherant is typically also a lubricant with some glidant properties as well. The differentiation between these terms is as follows:-

LUBRICANTS:-

- are intended to **reduce the friction** during tablet ejection between *walls of the tablet and the wall of the die cavity* in which the tablet was formed.
- **Example:-** Magnesium stearate, Calcium stearate, stearic acid, talc...

Lubricants...

- Lubricants may have undesirable effect:
 - ✓ May reduce tablet strength
 - ✓ Tablet disintegration & dissolution are often retarded
 - ✓ All these effects are related to the *amount of lubricant* used, the *duration* and *intensity* of mixing
- To avoid negative effects of the above lubricants, hydrophilic lubricants can be used

E.g., Surfactants, PEGs, or combined with hydrophobic

Lubricants...

ANTIADHERANTS:-

- are substances that function to *reduce adhesion* between the *powder* and *the punch faces* and thus prevent particles sticking to the punches in the process of tablet manufacturing.
- Reduce *sticking* or *picking* which is affected by the *moisture content* of the powder
- Adherence is especially prone to happen if the tablet punches are *engraved* or *embossed*
- Adherence can lead to a **build-up** of a thin layer of powder on the punches
- **Examples:-** *talca, Mg stearate, starch.....*

Lubricants...

GLIDANTS:-

- are intended to *promote flow* of the tablet granulation or powder materials by *reducing friction between the particles*.
- **Adequate flow** is essential in tableting
- Addition of such flow promoters is critical specially during *high production speeds*
- **Examples:-** silica, talc, corn starch.....

Coloring Agents

- ❖ These are substances used to *impart color to drugs* so as to *increase their organoleptic properties*.
- ❖ Can be added as an insoluble powder or dissolved in the granulation liquid
- ❖ The use of colors in tablet making served three purposes:-
 - ✓ Disguising of off-color drug
 - ✓ Product identifications
 - ✓ Production of more elegant production



Flavouring Agents

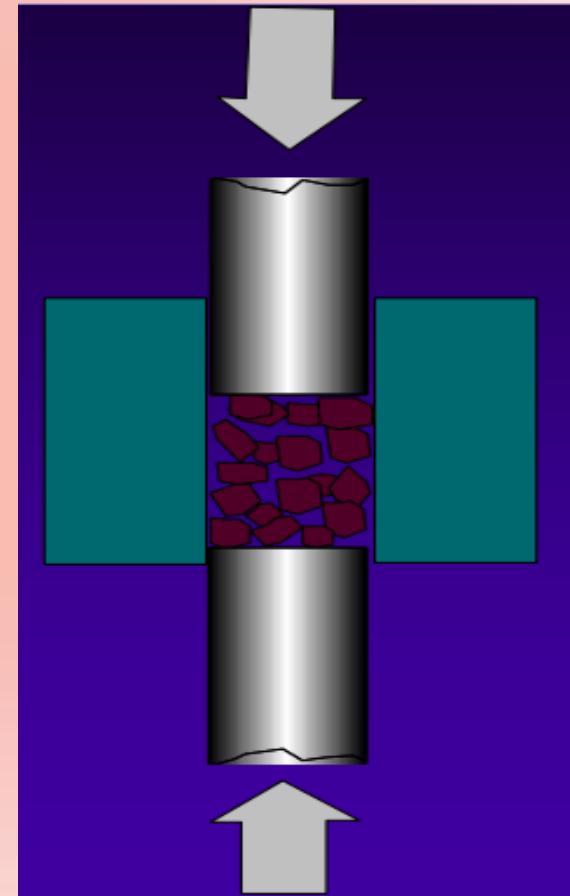
- These are the substances which are used to impart flavor to the preparation
- Generally, water-soluble flavors have poor stability; hence, flavor oils or dry powders usually are used
- Usually, the maximum amount of oil that can be added to a granulation without influencing its tablet characteristics is ***0.5% - 0.75%***

Sweetening Agents

- These are substances which are used to mask the bitter taste of the drug and to make the preparation sweet in taste
- Like flavors, are usually used with chewable tablets or tablets dissolve in the mouth
- Some sweetness may come from the diluent (e.g., mannitol, lactose); agents, such as *saccharin* and *aspartame*, can also be added

Tablet Manufacturing Methods

- Tablets are made by compressing a particulate solid b/n *two punches in a die* of a tablet press
- Formulation must have *three* essential attributes to be transformed into quality tablets:
 1. Must *flow* into the die of the tablet press *sufficiently rapidly* and in a *reproducible* manner → to avoid *content* and *weight variation*

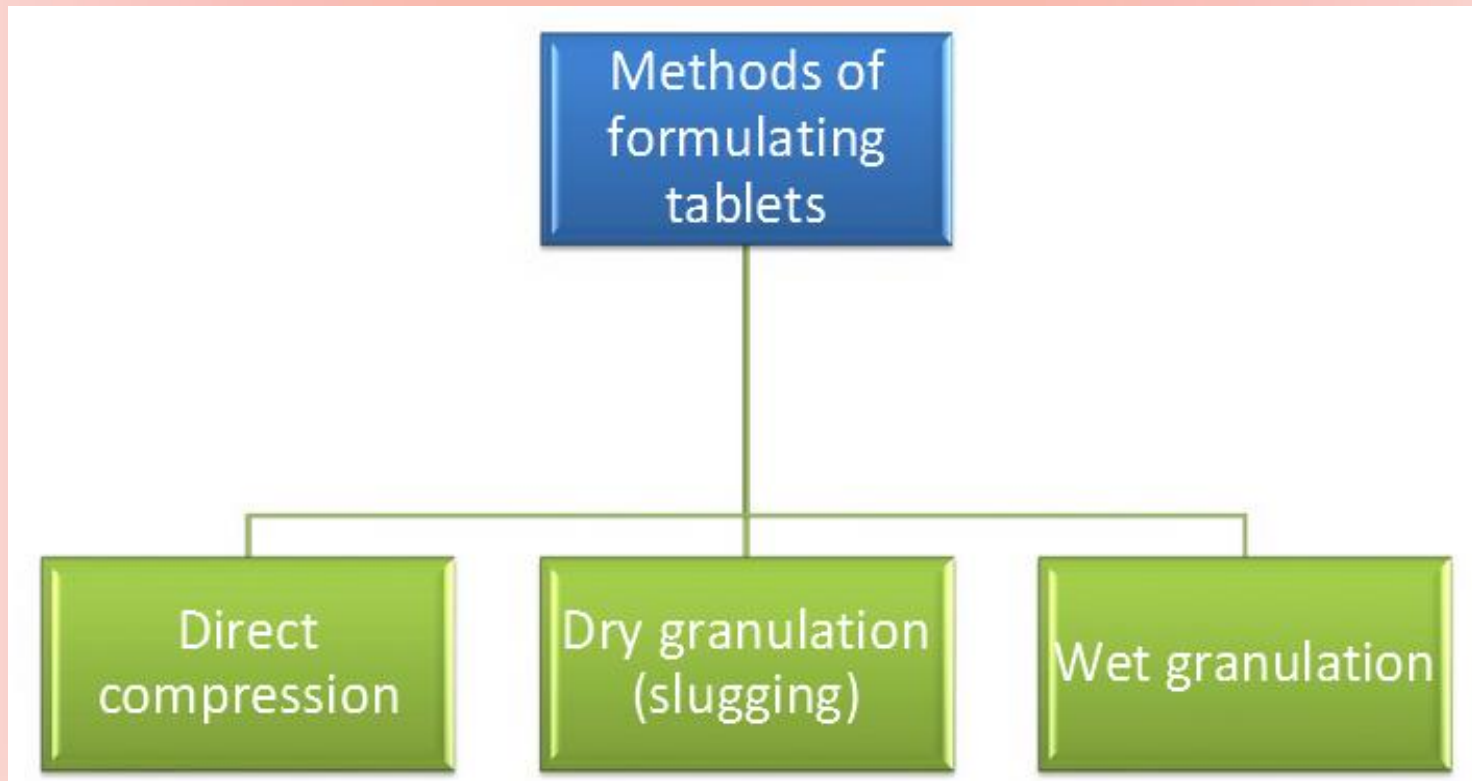


Tablet Manufacturing...

2. Particles of the formulation must *cohere* when subjected to a compressing force and *remain coherent* after removal of force
3. After the compression event is complete, it must be possible for the tablet to be *removed* from the press *without damage* to either the tablet or the press

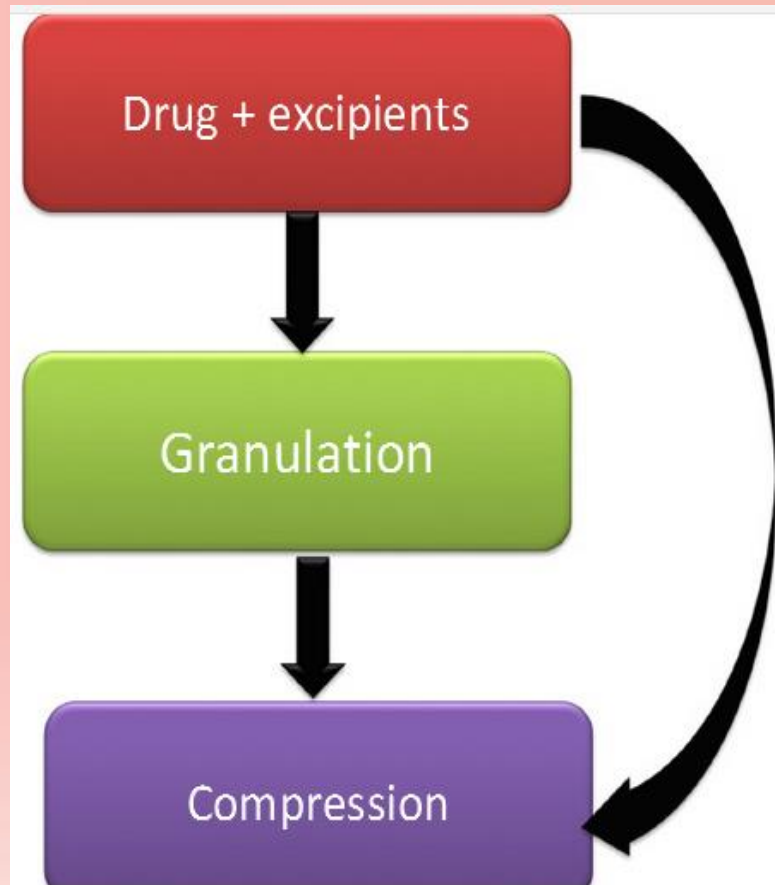
Tablet Manufacturing...

- There are three methods of tablet manufacturing:



Tablet Manufacturing...

➤ Methods of tablet manufacturing:

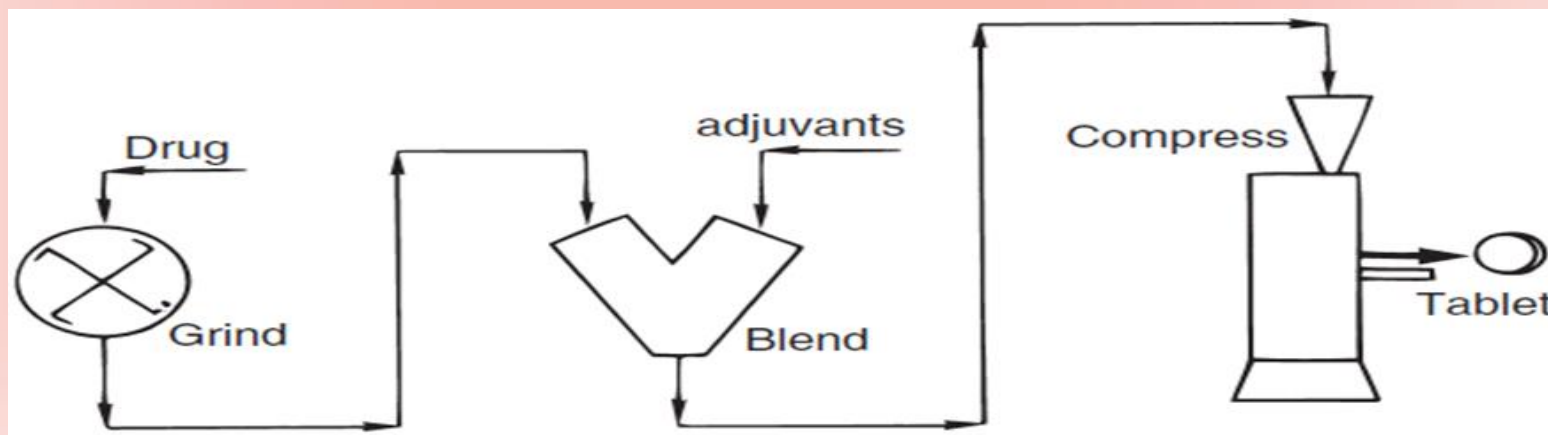


Direct Compression

- It is a method of tablet manufacturing in which:-
 - materials possessing *good flow* and *compressibility* undergoes prior to granulation to give tablets.
 - powdered drugs are combined with a pregranulated
- *Granular/crystalline* chemicals like KCl possess **free-flowing** and **cohesive** properties that enable them to be compressed directly → no need of granulation
- **Examples:** *Spray-dried lactose*, *alpha lactose monohydrate*, *MCC*, & *dicalcium phosphate*

Direct Compression...

- *Most large dose drugs do not lend themselves to this process*
- With many other drugs having **small doses**, uniform blends of the drug and coarser direct compression diluents can not be achieved which makes the process impractical.
- However, the use of compressible diluents with many **moderate-dose** drugs makes the process the most stream lined method of tablet manufacturing.



Direct Compression...

Advantages:

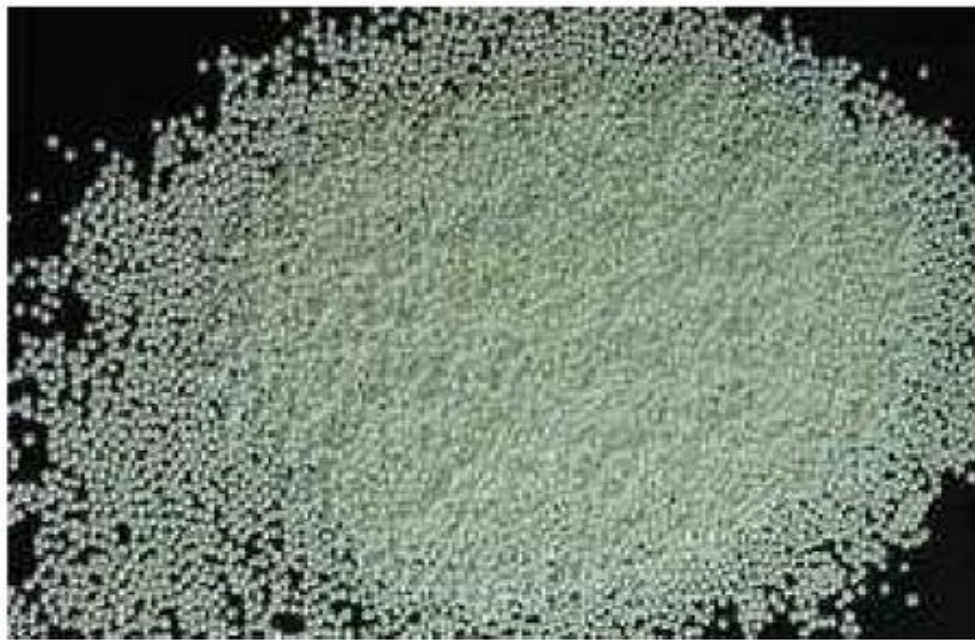
- low labor input
- it is a dry process
- it involves fewest processing steps

Limitations:

- poor content uniformity of the drug in the compressed tablet
- tablets **softer** than those produced by wet granulation
- not employed for most large-dose drug preparation
- static charge build up

Dry Granulation

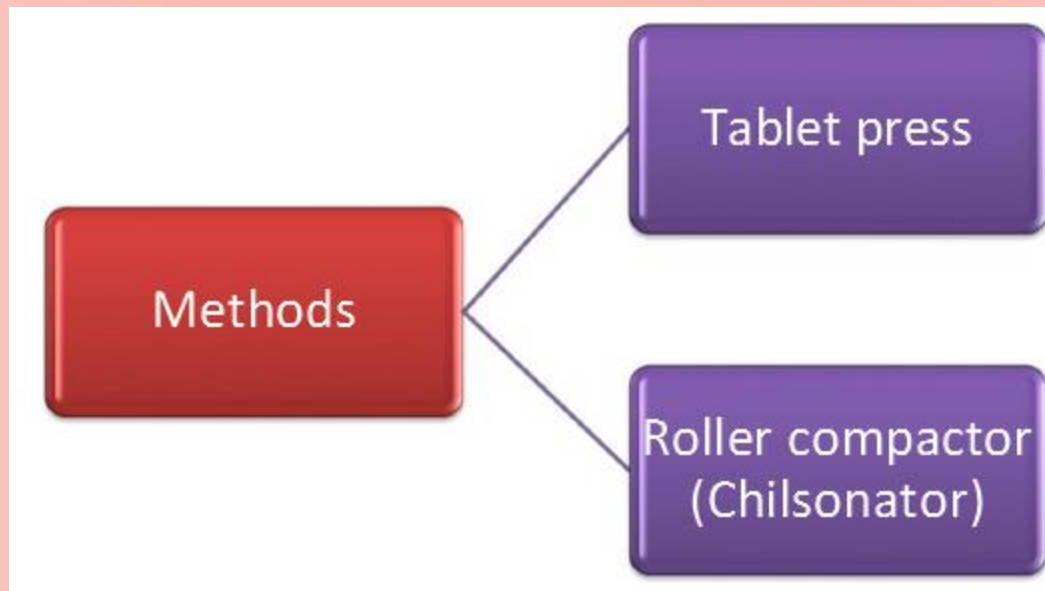
- It is a valuable technique in situations where:-
 - the effective dose of a drug is too high for direct compression
 - the drug is sensitive to heat, and moisture



Granulation

Dry Granulation...

- It involves two methods:



- The compaction of the components of a tablet formulation is followed by milling and screening prior to final compression into a tablet.

Dry Granulation...

A) Tablet Press/Slugging:

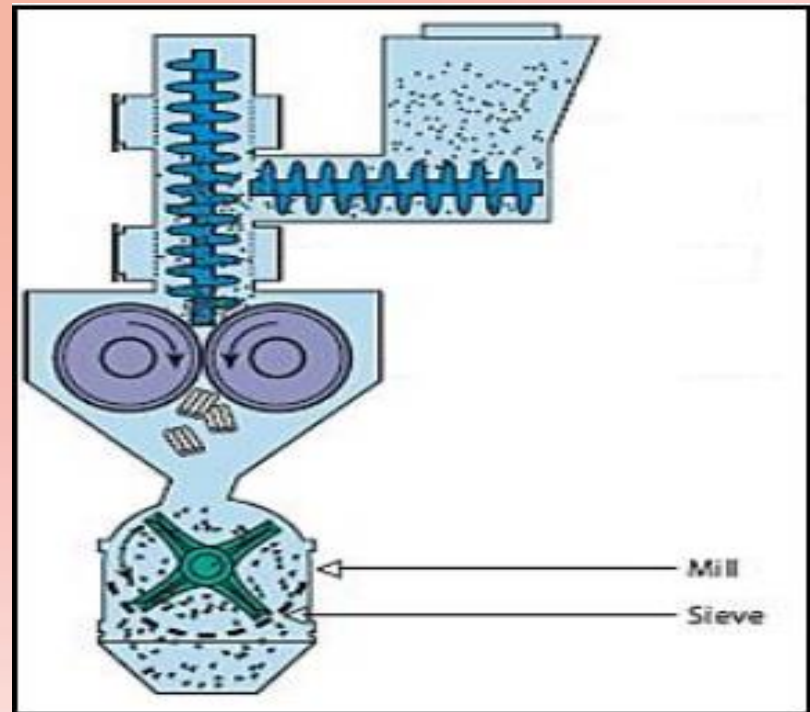
- ***Large tablet*** produced by using conventional tablet press
- Tablets produced at this stage (***slugs***) will not be of acceptable quality
- Tablet is then ***milled*** to produce granules of required size
- After ***sieving***, can then be ***compressed again*** to give satisfactory tablets

i.e. Powdered material → *slug* → *granule*

Dry Granulation...

B) Roller Compaction:

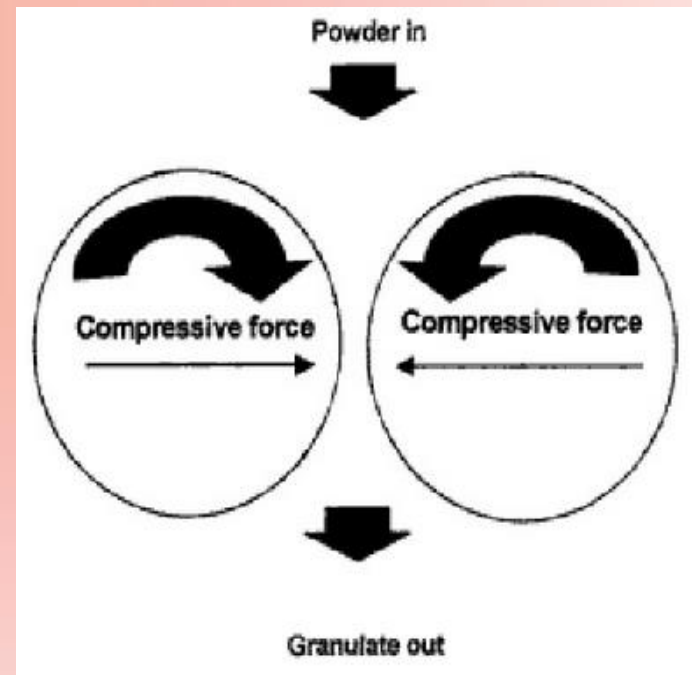
- Powder mixture passed between two *counter-rotating cylindrical rollers* to form *sheet/film* of compressed material
- **Broken** down to a product of granular size and then **recompressed**



Dry Granulation...

Working of roller compactor:

- The basic concept of compaction is to force fine powders b/n two counter rotating rolls.
- **Compression:** stress from the rollers compresses the powder mass into compacts
- **Milling:** compacts are milled to form small granules
- **Screening:** the granules are screened to select desired size range



Dry Granulation...

Advantages:

- Useful for moisture sensitive drugs
- Fewer steps and equipment than wet granulation
- Less loss of materials e.g., transfer steps (mixer-granulators)
- May be less expensive
- Less time-consuming

Dry Granulation...

Disadvantages:

- Lots of force used to compress the dry mixture
- Makes very hard granules—disintegration/dissolution can be affected
- May generate static charge
- The process tends to create more dust than wet granulation, increasing the potential contamination

Wet Granulation

- Pretreatment that is usually necessary to make powders take the form of granulation
- *A widely employed method for the production of compressed tablets*
- It serves several purposes in the tablet manufacturing process:
 1. Improves flow by forming spherical particles with large particle size
 2. Improves compression characteristics → ↑cohesive strength
 3. Homogeneous mixture is achieved → segregation is prevented
 4. It reduces dust

Wet Granulation...

- The unique portions of wet granulation process involve the *wet massing* of the powders, *wet sizing*, and *drying*.
- Wet granules formed by forcing the wet mass through a *sieve*
- Solvent usually contains a dissolved binder, which holds the granule together after drying
- After drying, the granules are *screened* to produce appropriately sized particles

Wet Granulation...

Advantages:

- Works well for many drugs
- Useful for fluffy powders that do not flow well or mix well
- **Content uniformity** in tablets can be increased for low dose drugs
- Lower pressures are needed to compress tablets

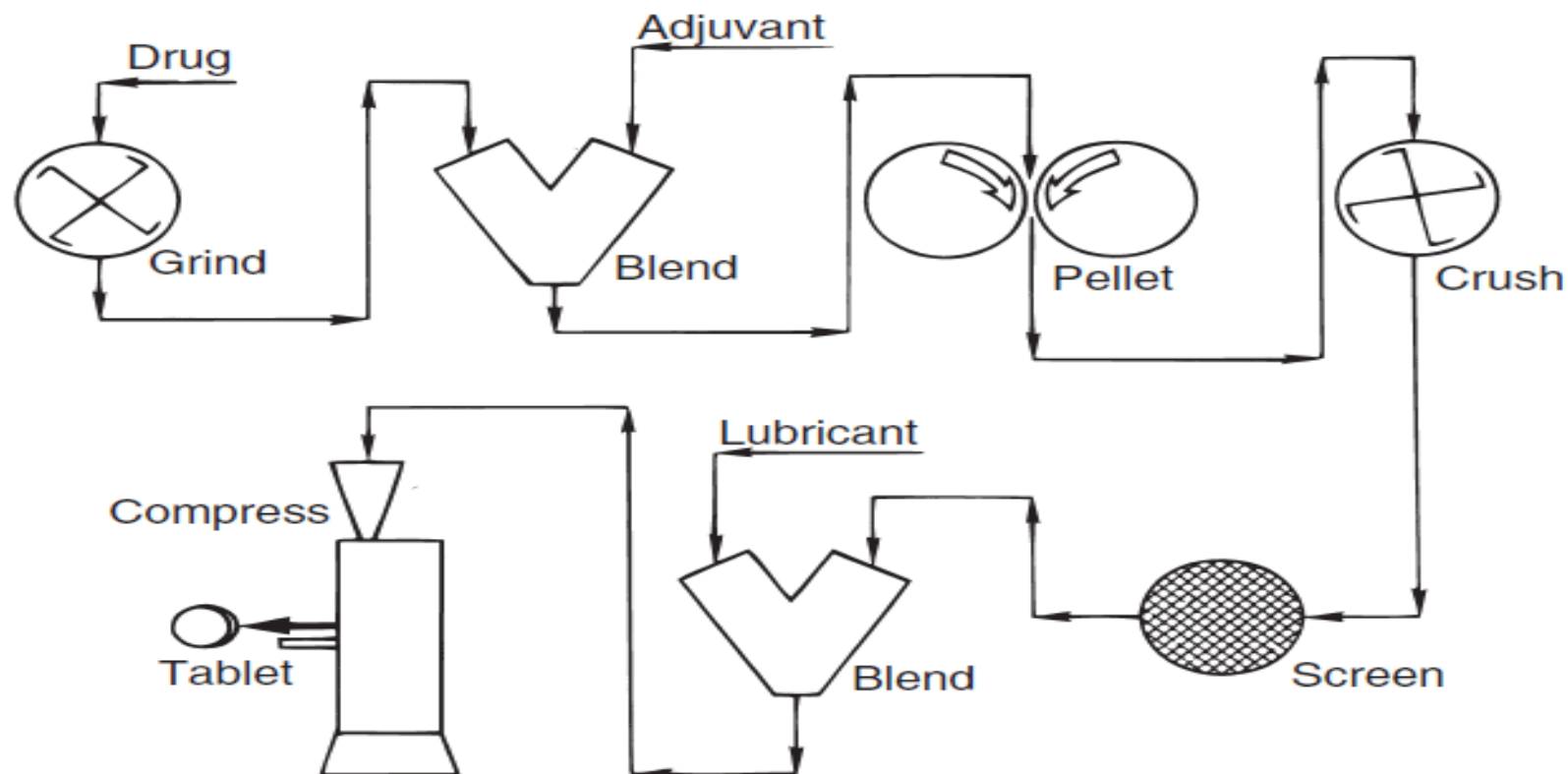
Wet Granulation...

Disadvantages:

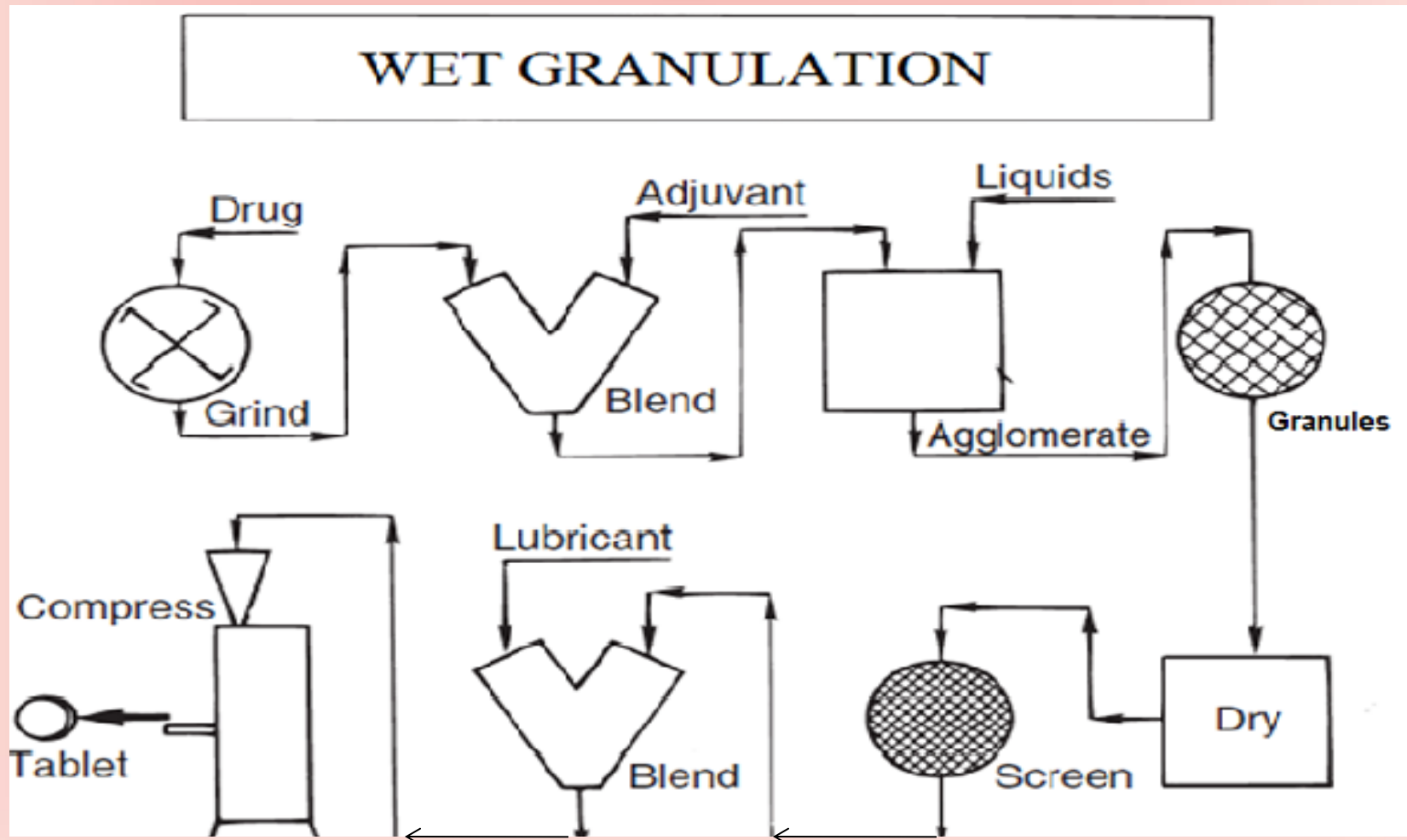
- Stability may be major concern for moisture sensitive or thermolabile products
- **Multi-step** process: requires lots of equipment, longer time, large space, many personnel,
- Has inevitable *materials loss*
- Multiple processing steps add complexity and make validation and control difficult

Operation Processes in Granulation

DRY GRANULATION



Operation Processes in Granulation...



Granule Characteristics

- The characteristics of a tablet that makes it a popular DF are in general dictated primarily by the qualities of the granulation from which it is made.
- Basically, materials intended for compaction into a tablet must possess **two characteristics:-**

I. FLUIDITY(FLOW)

- Good flow properties are essential for the transport of the material through the hopper, into and through the feed frame, and into the dies.
- Tablet material should therefore be in a physical form that flows smoothly and uniformly.
- *Spherical shape* offer minimum contact surfaces between themselves and the walls of the machine parts.

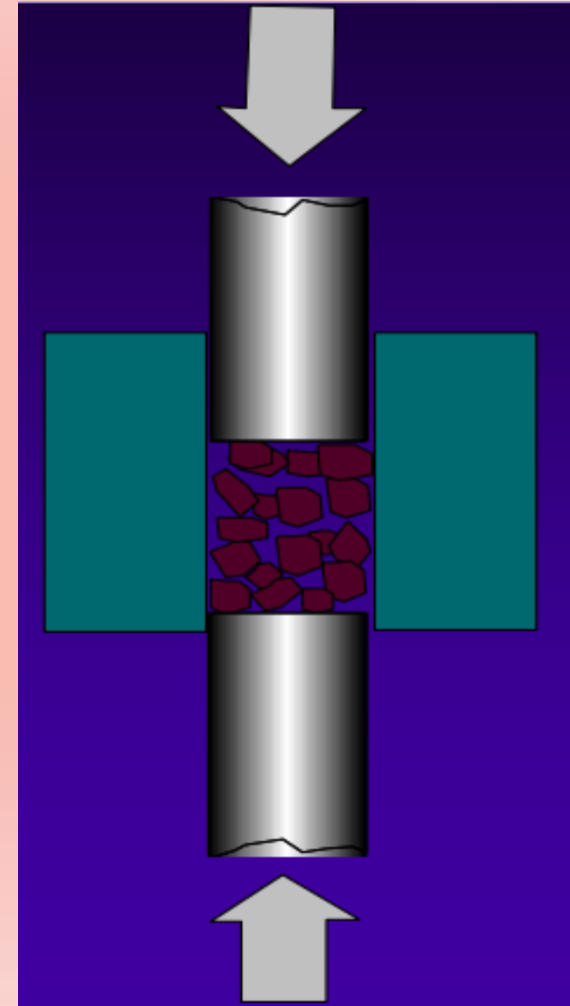
Granule Characteristics...

II. COMPRESSIBILITY

- This is the property of forming a stable compact mass when pressure is applied.
- Tableting is a **compaction process**, a general term used to describe a situation in which powdered materials are subjected to some level of mechanical force and may be simply stated as *compression* and *consolidation* of a two phase system due to the applied force.

Granule Characteristics...

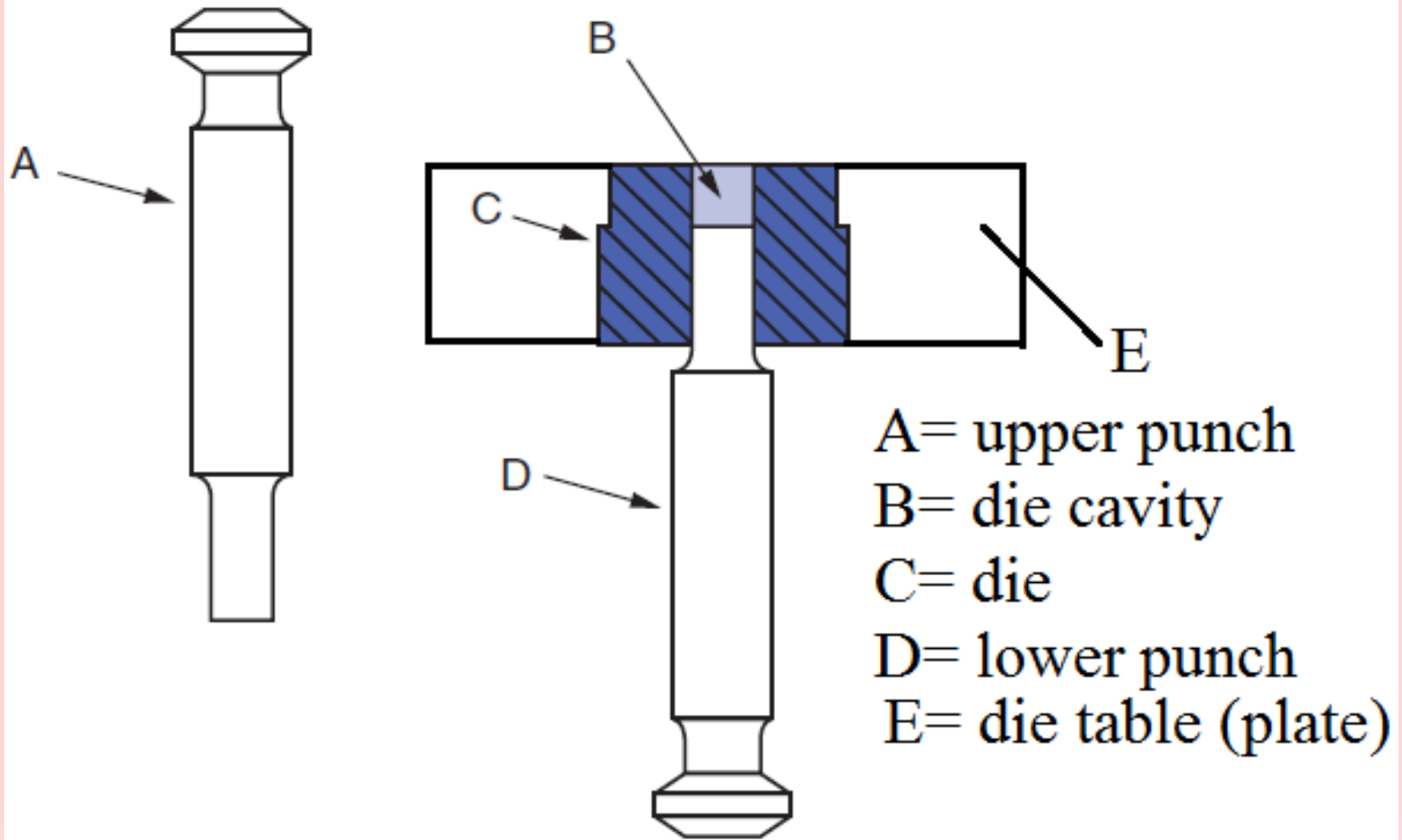
- **Compression**:- means a reduction in the bulk volume of the material as a result of displacement of the gaseous phase.
- **Consolidation**:- increased mechanical strength due to interparticulate interactions



Tablet Compression Machine and Tooling

- Tablets are made by compressing a formulation containing a drug or drugs with excipients on stamping machines called presses.
- Tablet compression machines (tablet presses) are designed with the following basic components:-
 1. **HOPPER:-** for holding and feeding granulation to be compressed.
 2. **DIES:-** that define the size and shape of tablet
 3. **PUNCHES:-** for compressing the granulation with in the dies.
 4. **CAM TRACKS:-** for guiding the movement of the punches
 5. **A FEEDING MACHINE:-** for moving granulation from the hopper into dies(hopper shoe)

Tablet Compression...



Tablet Compression...

- Two basic designs of tablet presses in pharmaceutical industry:

1. Single-punch presses (eccentric) and

2. Rotary presses (multistation)

- Each varying in *productivity* but similar in *basic function* and *operation*
- Tablet formulation compressed within a steel *die cavity* by the pressure exerted by the movement of **two** steel *punches*

Tablet Compression...



Tablet Compression...

Compression machine tooling

- Each tooling sets consists of a die and upper and lower punches.
- The *size and shape* of the tablet as well as certain *identification markings* are determined by the compression machine tooling.
- Since each tablet is formed by a tooling set, the tooling must meet many requirements to satisfy the needs of:
 - *dosage uniformity,*
 - *production efficiency, and*
 - *esthetic appearance.*

Tablet Compression...

Stages within the compression process

- Irrespective of the type of press used, there are **three** major stages of the compression process

Stage I: *Filling stage*

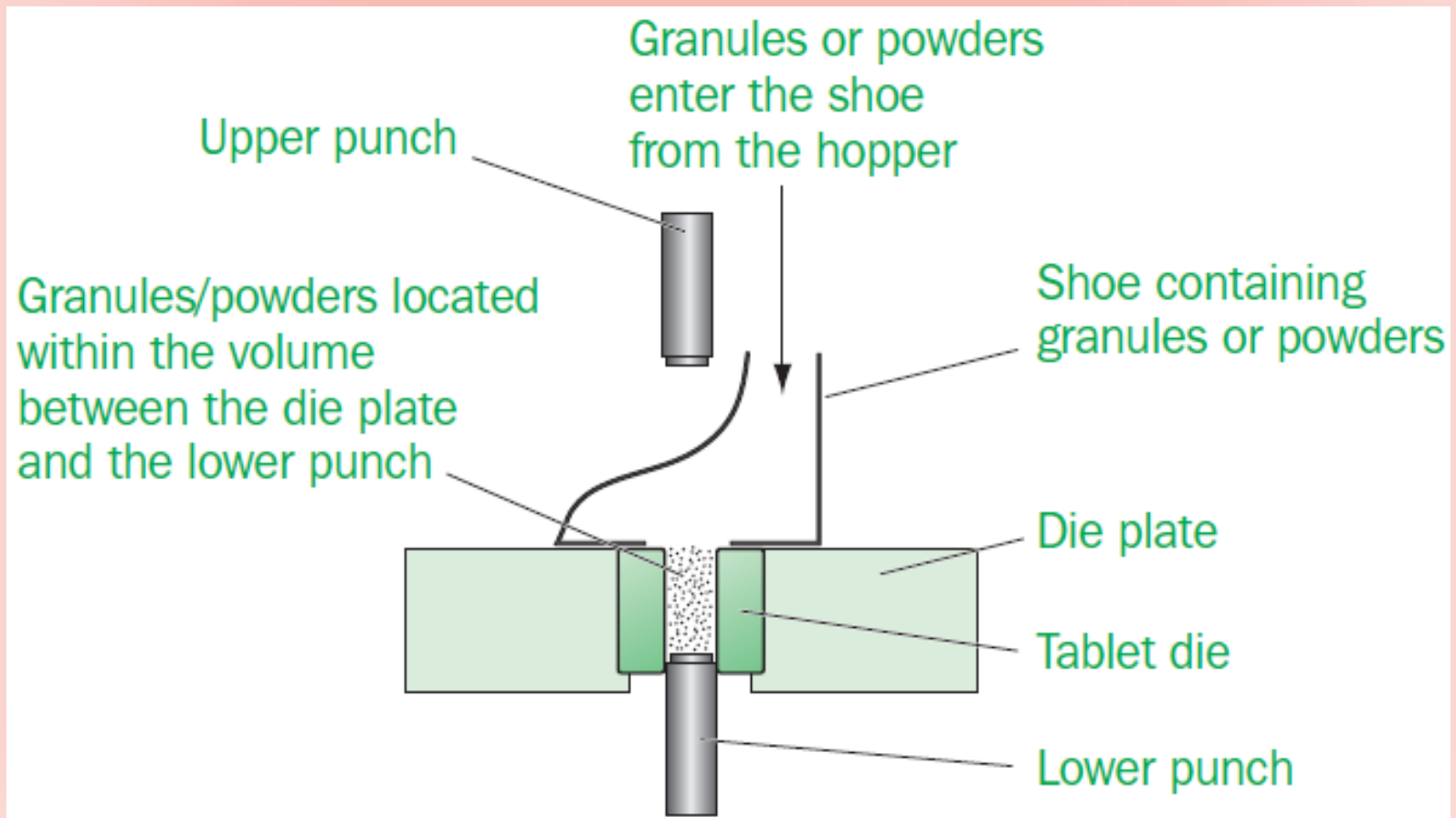
- Lower punch falls within the die → forming a cavity
- Upper punch remains suspended
- Powders and granules flow under the influence of gravity from a hopper (hopper shoe over the die cavity)

Tablet Compression...

- Though tablets are usually described in terms of weight, the die is filled by a volumetric process
- Volume of space occupied by the granules/powders is defined by the *position of the lower punch*
- This volume should be filled reproducibly on each occasion
→ uniform mass of tablets, uniform content

Tablet Compression...

Filling:



Tablet Compression...

Stage II: *Compression stage*

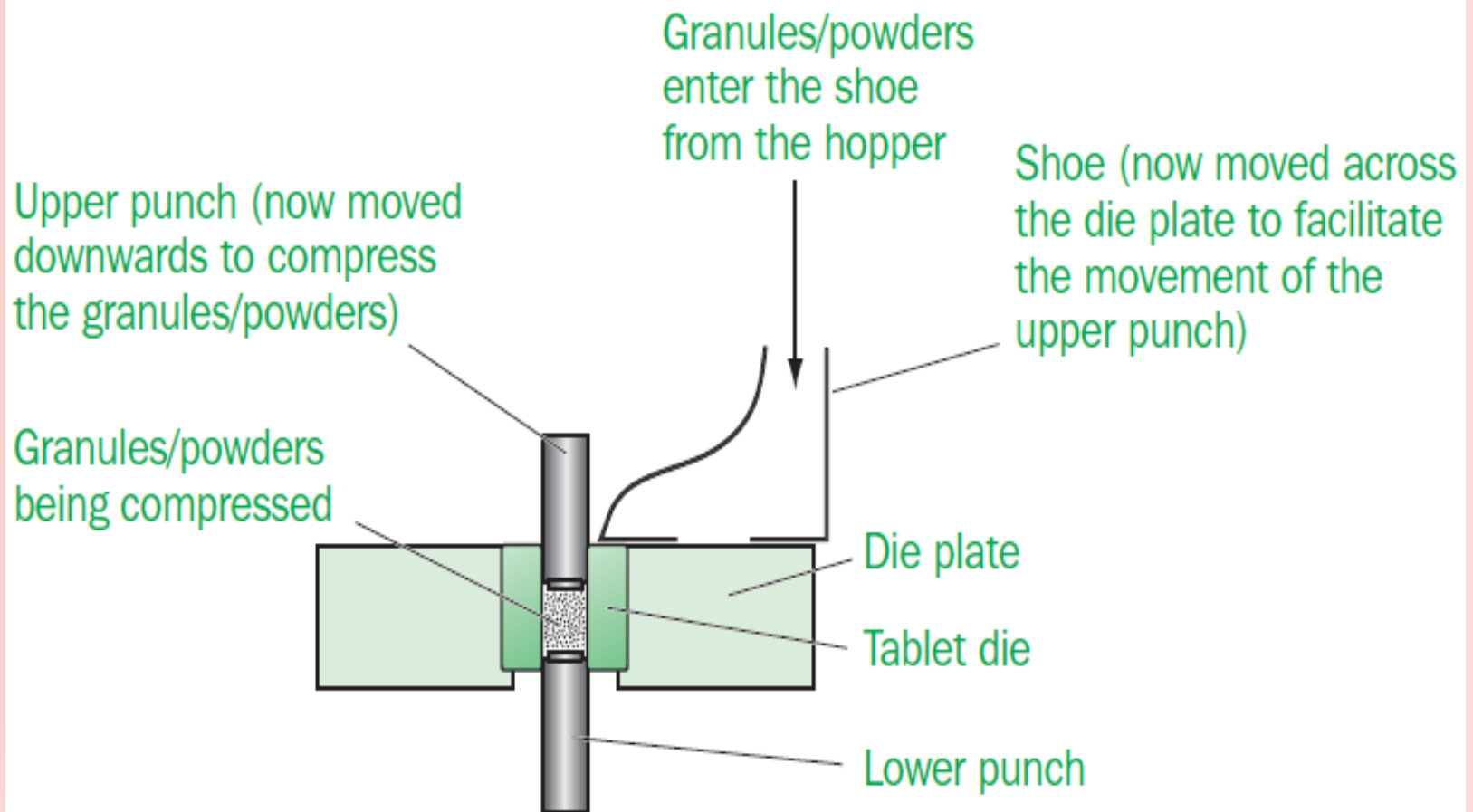
- Feed shoe moves back (single-punch press)
- Upper punch descends, and its tip enters the die, confining the particles
- Distance between the punch faces decreases, either by movement of the *upper punch alone* (as in eccentric presses) or by movement of *both punches* (as happens in rotary presses)

Tablet Compression...

- Porosity of the contents of the die is progressively reduced, and the particles are forced into ever-closer proximity to each other
- Interparticulate forces then cause the individual particles to aggregate, forming a tablet
- Magnitude of the force is governed by the minimum *distance separating the punch faces*

Tablet Compression...

Compression:



Tablet Compression...

Stage III: *Ejection stage*

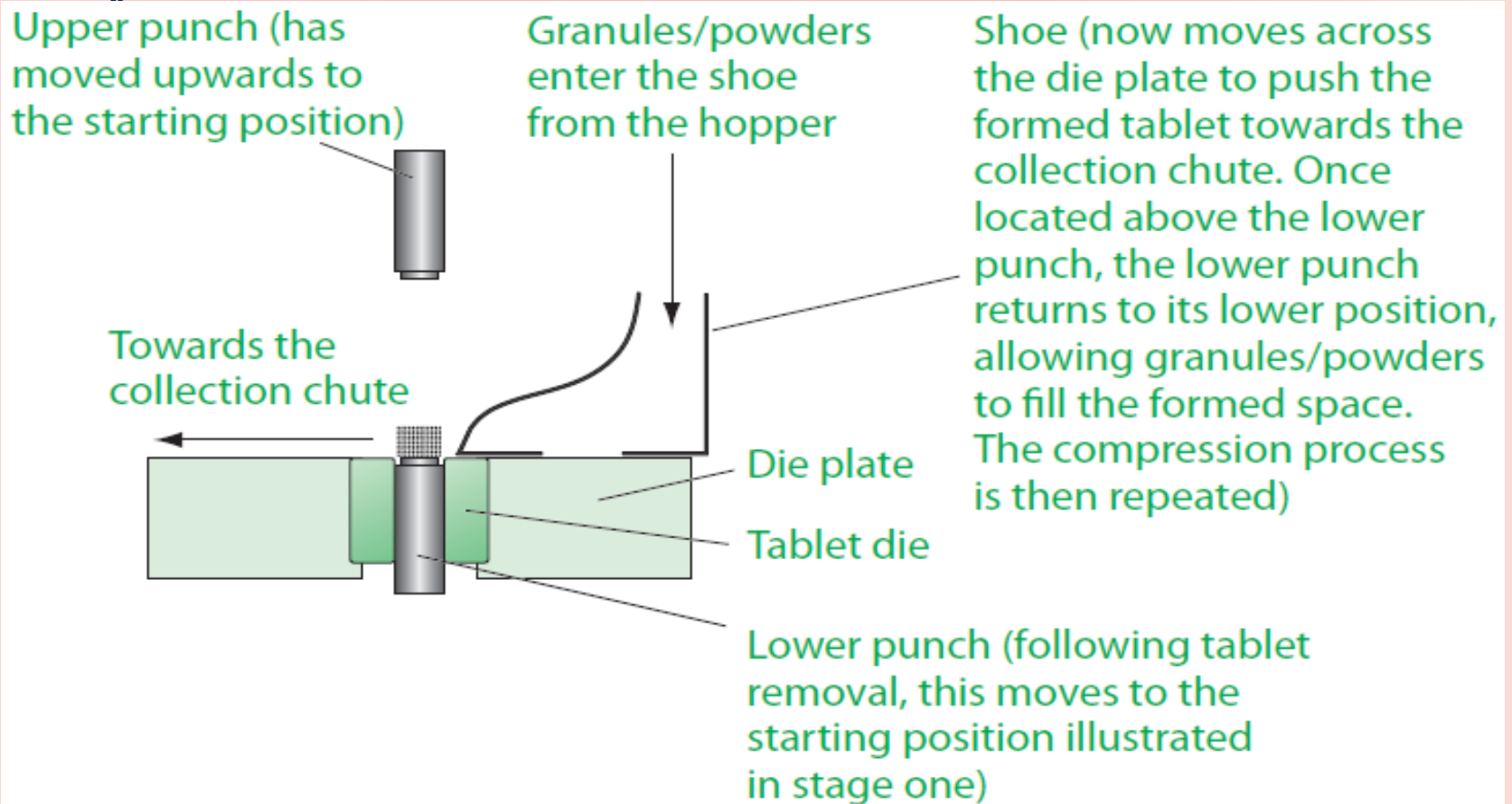
- Upper punch is withdrawn from the die → force being applied to the tablet is removed
- This would result in a **decrease** in interparticulate contact and hence tablet strength
- As the upper punch leaves the die, the lower punch moves upwards, pushing the tablet

Tablet Compression...

- The shoe of the hopper is then moved across the die plate
→ it pushes the tablet from the lower punch
- Simultaneously the lower punch descends to its starting position and granules/powders fill the volume between the shoe and the lower punch
- *The manufacturing sequence returns to stage 1 once more*

Tablet Compression...

Ejection:



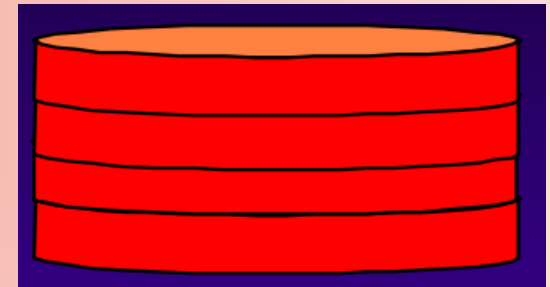
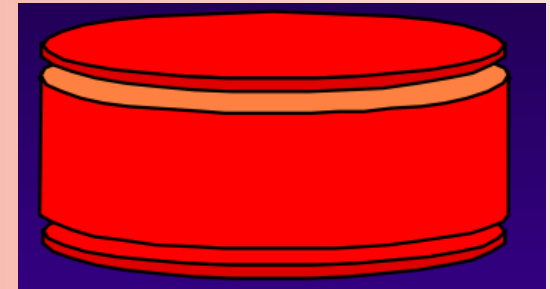
Tablet Processing Problems

- ❖ Various problems can occur during the manufacturing of tablets. The source of the problem could be:-
 - *The formulation*
 - *The compression equipment*
 - Both
- ❖ Formulation possessing adequate technical properties and suitable, well conditioned tablet press should be used
- ❖ Some of the tablet processing problems are:-

Tablet Processing Problems...

1. *Capping and Lamination*

- ❖ **CAPPING**:- is the partial or complete separation of the top or bottom crowns of a tablet from the main body the tablet.
- ❖ **LAMINATION**:-is the separation of a tablet into two or more distinct layers.
- ❖ Usually these problems are readily apparent immediately after compression, however the problem may occur hours or even days later.



Tablet Processing Problems...

Causes of capping and lamination

- ❖ Air entrapped in the granulation expands after compression
- ❖ Due to large amount of fines
- ❖ Too much or too little lubricant or excessive moisture
- ❖ The tooling (if the geometry of the punches is intricate (e.g., curved face))
- ❖ Lower punch remains below die surface during ejection

Tablet Processing Problems...

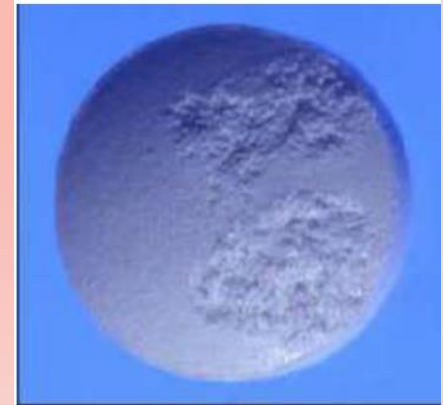
Possible solutions

- ❖ Pre-compression
- ❖ Reducing the final compression pressure
- ❖ Use of flat punches
- ❖ Optimum level of moisture
- ❖ Adjustment of the set up at the press

Tablet Processing Problems...

2. *Picking and Sticking*

- ❖ **PICKING**:- the surface material from a tablet that is sticking to and being removed from the tablets surface by punch.
 - It is a particular concern when punch tips have engraving or embossing.
 - Small enclosed areas such as those found in the letters 'B', 'A', 'O' are difficult to manufacture cleanly.
- ❖ **STICKING**:- refers to tablet material adhering to the die wall.



Tablet Processing Problems...

Possible solutions

- ❖ Increasing the concentration or changing of lubricant
- ❖ Decreasing moisture content of granulation (proper drying)
- ❖ By altering the mixing conditions (time and rate)
- ❖ Punches should be *polished, cleaned, or coated* regularly
- ❖ Letters should be designed *as large as possible*
- ❖ Use of high MP excipients

Tablet Processing Problems...

3. *Mottling*

- ❖ It is unequal distribution of color on a tablet.
- ❖ Color non-uniformity with in a single tablet, tablet to tablet or between lots to lots.

Causes:

- a drug whose color differs from the tablet excipients
- a drug whose degradation products are colored
- the solvent system
- improperly mixed dye, especially during DC
- a dye migrates to the surface of granulation while drying



Tablet Processing Problems...

Possible solutions

- ❖ Use of colorants
- ❖ Reducing the drying temperature
- ❖ Grinding to a smaller particle size



Tablet Processing Problems...

4. Weight Variation

- ❖ Weight is determined by the amount of granulation in the die prior to compression
- ❖ Anything that can alter the die-filling process can alter tablet weight

Possible causes:

- ❖ Variation in granular size & size distribution
- ❖ Poor flow of granules
- ❖ Punch variation - when punches are not of equal length

Tablet Processing Problems...

Possible solutions

- ❖ Maintain uniform size & distribution of granules
- ❖ Addition / increase the amount of glidant
- ❖ Use of induced die feeders (mechanically forcing granules)
- ❖ Use punches of equal length
- ❖ Optimum mixing time

Evaluation of Tablets

- ❖ Prepared tablets must meet *physical specification* and *quality standards* according to the monograph of the pharmacopoeia
 - thickness,
 - hardness,
 - friability,
 - weight variation,
 - content uniformity,
 - disintegration, and
 - dissolution
- ❖ These factors must be controlled during tablet production (*in-process control*) and are validated *after the production* to ensure the quality standards

Evaluation of Tablets...

General Appearance

- ❖ Refers a tablet's *visual identity* and over all “*elegance*”
- ❖ Is essential for:
 - Consumer acceptance
 - Control of lot-to-lot and tablet to tablet uniformity
 - Monitoring trouble free manufacturing
- ❖ The control of the general appearance of a tablet involves the measurement of a number of attributes such as:-
 - Size, shape, color, odor, taste, texture, physical flaws and consistency, legibility of identifying markings

Evaluation of Tablets...

Size and shape:

- ❖ Tablets shape and dimensions are determined by the *tooling* during compression
- ❖ **Thickness** of the tablet is dimensional variable related to the process
- ❖ At a constant compressive load tablet thickness varies with:
 - ☛ Change in die fill
 - ☛ Particle size distribution and packing of the particle mix
 - ☛ Tablet weight

Evaluation of Tablets...

- ❖ At a constant die fill, thickness varies with variation in compressive load
- ❖ Tablet thickness is consistent batch to batch or with in a batch only if:-
 - The tablet granulation or powder blend is adequately consistent in particle size & size distribution
 - If the punch tooling is of consistent length
 - The tables press is clean and in good working order

Evaluation of Tablets...

- ❖ **Thickness** of tablets may be measured by :-
 - Micrometer (accurate measurement for individual tablets)
 - Sliding caliper scale involves placing 5 or 10 tabs in a holding tray where their total crown thickness may be measured with a sliding caliper scale.
- ❖ tablet thickness should be controlled within a $\pm 5\%$ variation of a standard value.



Evaluation of Tablets...

Unique identification markings

- ❖ Pharmaceutical companies manufacturing tablets often use some type of unique markings on the tablet in addition to color, to aid in the rapid identification of their products.
- ❖ The type of informational markings placed on a tablet usually includes:-
 - *The company name or symbol*
 - *Product code number*
 - *The product name*
 - *Product potency*

Evaluation of Tablets...

Organoleptic properties

- ❖ These involve evaluation of color, odor, taste, and level of flaws.
- ❖ **COLOR:-** many tablet manufacturing pharmaceuticals use color as a vital means of *rapid identification* and *consumer acceptance*.
- ❖ The color of the product must be uniform with in:
 - a single tablet,
 - from tablet to tablet and
 - from lot to lot.

Evaluation of Tablets...

- ❖ **ODOR:-** the presence of an odor in a batch of tablets could indicate stability problem.
- ❖ However, the presence of an odor:-
 - Could be characteristics of the drug(e.g. vitamins), added ingredients
 - Could be the DFs(e.g. film-coated tab)
- ❖ **TASTE:-** the control of taste is important in case of production of chewable tablets.

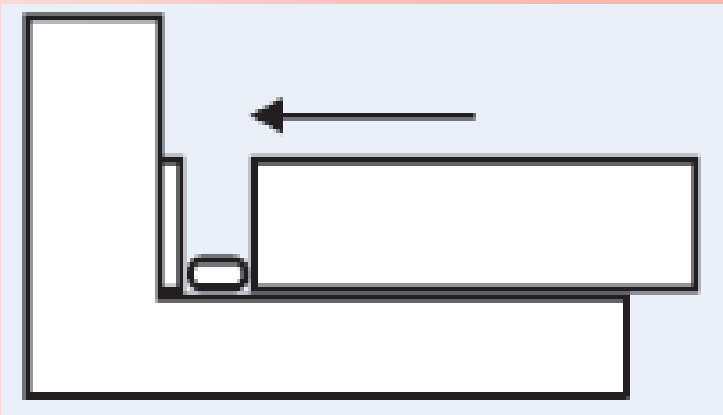
Evaluation of Tablets...

Hardness or Breaking Strength

- ❖ Tablets require a certain amount of strength, to withstand **mechanical shocks** of handling in manufacture, packaging, and shipping
- ❖ Necessary requisite for consumer **acceptance**
- ❖ Defined as the force required to break a tablet in a *diametric* compression test
- ❖ Tablets should be sufficiently hard to resist breakage but have **no problem in disintegration**
- ❖ The hardness of the tablet, like its thickness, is a function of the **die fill** and **compression force**.

Evaluation of Tablets...

- ❖ Tablet is placed between two anvils (one stationary and the other movable), and the crushing strength that just causes the tablet to break is recorded
- ❖ Generally, the greater the pressure applied, the harder the tablets, although the formulation composition and manufacturing process may also change tablet hardness



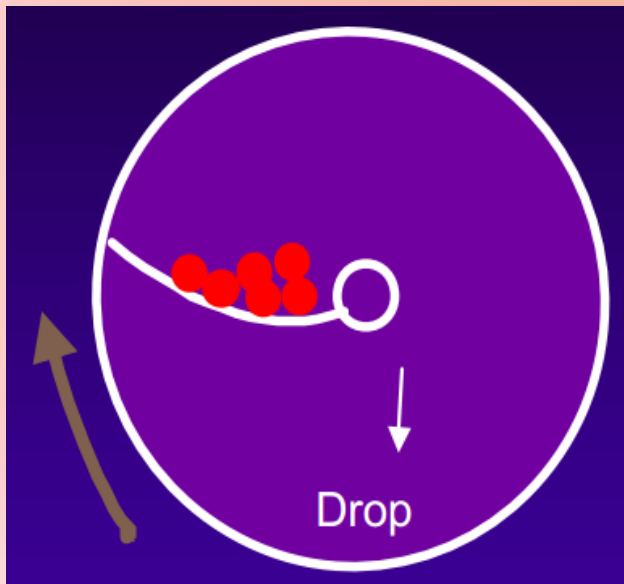
Evaluation of Tablets...

Friability

- ❖ *Hardness is not an absolute* indicator of tablet strength since some formulations tend to “cap” on attrition, losing their crown portions
- ❖ Another measure of a tablet’s strength is tablet *friability*
- ❖ High friability causes *lack of elegance* and *consumer acceptance* and even *weight variation or content uniformity* problems, create dirty processes during *coating & packaging*

Evaluation of Tablets...

- ❖ Pre-weighed tablet sample is placed in the friabilator → then rotated for 100 revolutions → The tablets are then dusted and reweighed → loss in weight is determined as percentage
- ❖ Compressed tablets that lose a maximum of **not more than 1%** of their weight are generally considered acceptable



Evaluation of Tablets...

Weight Variation

- ❖ The weight of the tablet being made is routinely measured to help ensure that the tablet contains the proper amount of drug
- ❖ **Quantity of fill** placed in the die of a tablet press determines the weight of the resulting tablet
- ❖ Depth of fill in the tablet die must be adjusted to hold a predetermined volume of powder or granulation
- ❖ During production, sample tablets are periodically removed for visual inspection and automated physical measurement known as **in - process control**

Evaluation of Tablets...

- ❖ According to USP, **20 uncoated tablets** are weighed individually and the average weight is calculated
- ❖ Compare individual tablet weights with the average
- ❖ Tablets meet the test if:
 - *no more than 2 tablets are outside the percentage limit and*
 - *no tablet differs by more than $2X$*
- ❖ Weight variation tolerances differ depending on average tablet weight

Evaluation of Tablets...

weight variation tolerances of uncoated tablets (USP)

Average tablet Wt (mg)	Tolerance allowed (%)
130 or less	10
130-324	7.5
More than 324	5

Evaluation of Tablets...

Content Uniformity

- ❖ Weight variation may be sufficient for determination of drug content uniformity
 - ❖ For tablets containing 90 - 95% API or
 - ❖ If the uniformity of the drug distribution in the granulation or powder from which the tablets were made were perfect.

Factors contributing to uniformity problems

1. Non-uniform distn of the drug s/ces throughout the powder mixture or granulation.
2. Segregation of the powder mixture or granulation during the various manufacturing processes.
3. Tablet weight variation

Evaluation of Tablets...

Test:

- 30 tablets are randomly selected → 10 of them are assayed individually

Result:

- 9 out of 10 must contain not less than 85% or not more than 115% of the labeled drug content
- The 10th tablet must not contain less than 75% or not more than 125%
- If these conditions are not met → remaining tablets from the 30 assayed individually → non falls out side of the 85 to 115% range

Evaluation of Tablets...

Disintegration

- ❖ For most tablets, the first important step toward solution is breaking down of the tablet into smaller particles or granules = **disintegration**
- ❖ The time that is required to disintegrate is measured in a device described in the USP.
- ❖ Uses 6 glass tubes that are 3 inches long, open at the top, and held against a 10-mesh screen at the bottom end of the ***basket rack assembly***

Evaluation of Tablets...

Test:

- ❖ One tablet is placed in each tube → the basket rack is positioned in a 1L beaker of water, simulated gastric fluid, or simulated intestinal fluid, at $37 \pm 2^\circ\text{C}$ → the basket assembly containing the tablets moves up and down

Result:

- ❖ The tablets must disintegrate, and all particles must pass through 10-mesh screen in the time specified

Evaluation of Tablets...



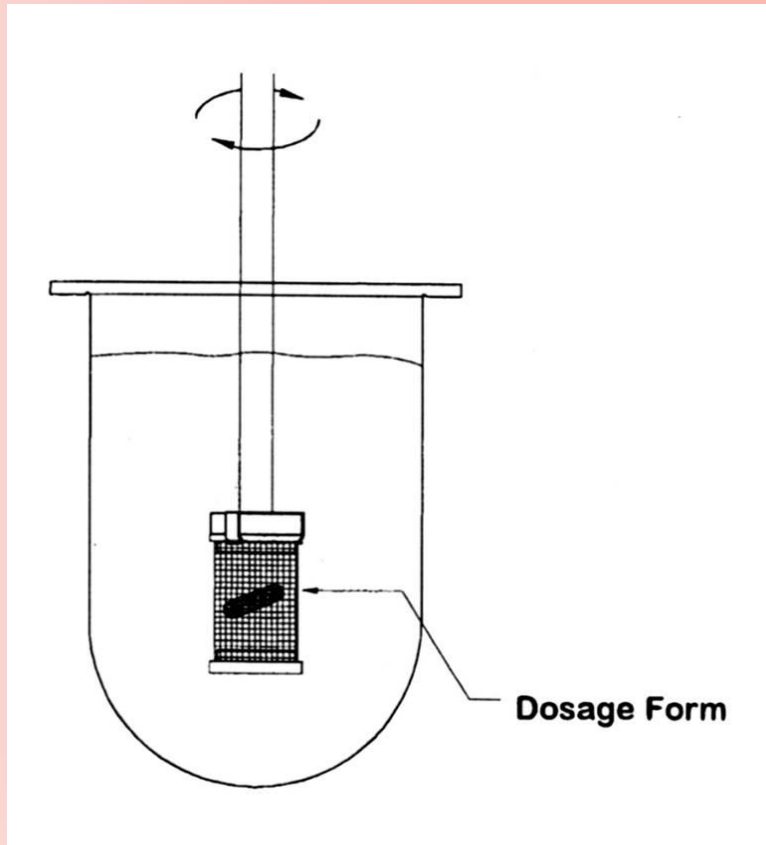
Evaluation of Tablets...

Dissolution Test:

- ❖ Disintegration test simply identifies the time required for the tablet to break under the conditions of the test
- ❖ It offers **no assurance** that the resultant particles will release the drug in solution at appropriate rate
- ❖ **Dissolution test** done to show the tablet disintegrated into particles release the drug in solution at appropriate rate
- ❖ Apparatuses used:
 - Basket apparatus (apparatus I)
 - Paddle apparatus (apparatus II)

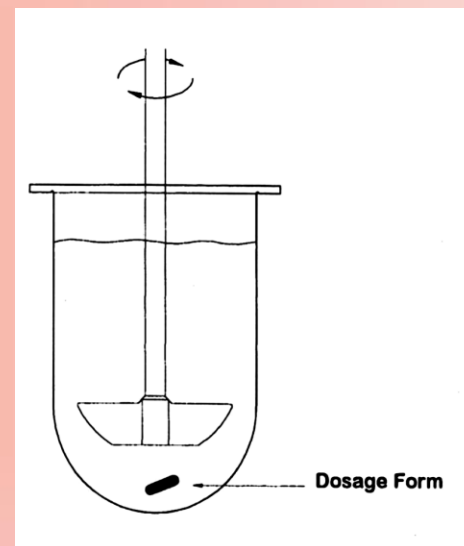
Evaluation of Tablets...

❖ Basket apparatus



Evaluation of Tablets...

❖ Paddle apparatus



Evaluation of Tablets...

Test:

- ❖ Different media at 37°C used → six tablets placed in apparatus
- ❖ Samples taken at time intervals
- ❖ Plot of concentration Vs time
- ❖ $t_{50\%}$, $t_{90\%}$ calculated
- ❖ Common dissolution tolerance in the USP for uncoated immediate release tablets is not less than 75% dissolved in 45min